

Jason Chen

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SUMMARY

Build games and animation worlds, and immersive XR applications. Weave new stories. Research the potential of artificial intelligence and agents for fields in computer graphics, computer vision, rendering, and 3D models.

EDUCATION

University of Pennsylvania, School of Engineering & Applied Science

May 2028

Candidate for a Bachelor of Science in Engineering in Digital Media Design. GPA: 3.93/4.00.

Philadelphia, PA

Relevant Coursework: Multivariable Calculus, Linear Algebra, Data Structures and Algorithms, Computer Systems, Computer Graphics, Computer Vision, 3-D Computer Modeling

SKILLS

Languages: Java, Python, C#, C++, C, HLSL, JavaScript/HTML/CSS

Engines and Software: Unity (Meta XR SDK), Unreal Engine, Blender (Python API), Maya, Rhino Grasshopper

Tools and Frameworks: Git, Perforce, Docker, CUDA, Photoshop, DaVinci Resolve

EXPERIENCE

Game Design Lead and Product Manager

August 2025 – Present

Penn UPGRADE (Game Research & Development)

Philadelphia, PA

- Leading scope, narrative, and game design for semester-long projects across a club of 20+ members
- Driving the organization of 5+ sub-teams of programmers, artists, and musicians for the development of a narrative rogue-like 2D game
- Shipped UpgradeKart, a cartoon-style multiplayer physics-based racing game for PC and console with 10 unique kart designs and multiple racetrack levels. Designed and programmed race karts, levels, and system UI.

VR Haptics Research and Development Intern

May 2026 – Aug 2026

RWTH Aachen Digital Additive Production

Aachen, Germany

- Developed an immersive VR lattice prototyping platform by integrating Rhino Grasshopper into Unity, enabling real-time parametric CAD design, haptic stiffness simulation, and cross-sections to accelerate additive manufacturing
- Optimized Grasshopper calculation and Unity import pipeline processing time from an average of 10 minutes to 2 minutes using data flow restructuring and serialization optimization.
- Edited the Meta XR SDK and SenseGlove source code in C# to enable more intuitive user interaction with the Meta Quest 3 headset and Nova 2 haptic gloves, respectively
- Coded and animated printing processes for a digital twin of the Aconity industrial LBPF 3D printer, used for operator training

Lead Engineer and Producer

June 2024 – August 2024

Carnegie Mellon University Game Academy

Pittsburgh, PA

- Led a team of 5 to develop an original VR cooking comedic horror game, Lulu's, in Unity 3D over two weeks
- Implemented finite state machines, AI behavior systems, and pathfinding algorithms to reduce latency, a frequent challenge in real-time VR applications
- Upgraded Meta's XR SDK to create novel player interactions including grabbing and pouring mechanics

PROJECTS

Procedural Terrain and Erosion Simulator | C++ , CUDA |

July 2026 – Present

- Developing a C++ terrain simulation implementing procedural heightmaps and hydraulic erosion to model terrain evolution.

Blender 2D Effects in 3D Animation Tool | Blender (Python API) |

January 2026 – May 2026

As a researcher under CGARP (Computer Graphics & Animation Research Projects)

- Conducted research and write documentation for submission to SIGGRAPH 2026
- Developed a custom tool using Blender's Python API to integrate 2D effects within 3D animation, streamlining the compositing pipeline for comic book style animations like Spider-Man: Into the Spider-Verse

INTERESTS

- Oil Painting, Photography, Ultimate Frisbee, Football, Basketball, Hiking, National Parks, Scuba Diving